1. Big-O Classification

Determine whether each of these functions is $O(x^2)$:

1) $f(x) = 17x + 11$
2) $f(x) = x^2 + 1000$
3) $f(x) = x \log x$
4) $f(x) = \frac{x^4}{2}$
5) $f(x) = 2^x$

2. Big-O Proof

Show that

$$\frac{x^3 + 2x}{2x + 1}$$

is $O(x^2)$.

3. Big-O Estimate for a Sum

Find a Big-O estimate for

$$\sum_{j=1}^{n} j(j+1).$$

4. Big-O Estimates

Give a Big-O estimate for each of these functions:

1) $f(n) = (n^2 + 8)(n + 1)$
2) $f(n) = (n \log n + n^2)(n^3 + 1)$
3) $f(n) = (n! + 2^n)(n^3 + \log(n^2 + 1))$

5. Bonus Task (extra 3 points)

Show that $n!$ is not $O(2^n)$.